

Ehsan Sharafian

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PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate developing intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate embedded hardware, IMU-based sensing, smartphone applications, biomechanics, machine learning, and haptic feedback for real-time gait analysis and personalized feedback. Proficient in Python, MATLAB, Android Studio/Java, PyTorch, and wearable system development, with six peer-reviewed journal publications and a strong focus on translating research into deployable medical and digital health technologies.

TECHNICAL SKILLS

Programming & Data Analysis	Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch
Mobile & Software	Matlab, Simulink, Android Studio, Git
Wearable & Embedded Systems	IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules, Altium Designer
Biomechanics & Signal Processing	Gait analysis, human activity recognition, sensor fusion, time-series analysis, feature extraction, OpenSim
Robotics	Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems, trajectory planning
CAD	SolidWorks

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant | [Biorobotics and Biomechanics Lab, University of Maine](#) Sep 2023 – Present
Orono, ME

- Designed and validated a real-time, smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories and classify level walking, stair ascent/descent, and ramp ascent/descent with 96.08% accuracy, less than 1.1 cm cumulative height error, and 112 ms delay.
- Engineered a lightweight human activity recognition system using two IMUs and machine learning techniques to detect normal locomotion with 99.1% accuracy and classify atypical stair-negotiation strategies in older adults with 93.1% accuracy, and 53 ms latency.
- Developed a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry to provide real-time thigh-extension feedback for adults age 65+, increasing stride length and walking speed by up to 30% in a single-session study with minimal cognitive demand.
- Validated wearable systems with human participants, including 40 adults age 65+ and 2 stroke patients in a haptic feedback gait study, 20 young adults in a foot-clearance study, and 10 young adults plus 15 older adults in activity recognition experiments.
- Implemented real-time algorithms and data analysis pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Tested and demonstrated the system with 6 experienced surgeons and supported medical-student workshops through operator training, troubleshooting, and usability feedback collection.

- Built experimental setups and prepared lab systems for robotics and mechatronics projects involving parallel manipulators, Delta robots, exoskeletons, CNC machines, 3D printers, CAD/MATLAB workflows, hardware setup, and technical documentation.
- Mentored 6 undergraduate students across 5 research teams, trained students on lab equipment and workflows, and supported technical organization for the 9th International Conference on Robotics and Mechatronics.

EDUCATION

Ph.D. in Mechanical Engineering University of Maine, Orono, ME	Expected Dec 2027
GPA: 4.00/4.00 Dissertation: Smart Real-Time Gait Training System	
M.S. in Mechanical Engineering, Robotics and Control Systems Amirkabir University of Technology (Tehran Polytechnic)	Feb 2020
GPA: 3.65/4.00 Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces	
B.S. in Mechanical Engineering University of Tehran	Jul 2017

SELECTED HONORS

Flagship Fellowship Award, University of Maine	Sep 2024 – Sep 2025
Graduate Student Government Award for conference proposal	Mar 2025

PUBLICATIONS

Six peer-reviewed journal publications in wearable sensing, biomechanics, rehabilitation engineering, and robotics.

1. Sharafian, E., & Hejrati, B. "Real-Time Foot Height Estimation and Activity Classification Using a Foot-Mounted IMU Implemented on a Smartphone." Sensors, 2026.
2. Sharafian, E., Ellis, C., & Hejrati, B. "Real-Time Activity Recognition Using Minimal Biomechanical Features: A Lightweight IMU-Based Classifier for Older Adults." IEEE Access, 2025.
3. Sharafian, E., et al. "The Effects of Real-Time Haptic Feedback on Gait and Cognitive Load in Older Adults." IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2025.
4. Noghani, M. A., Sharafian, E., Sidaway, B., & Hejrati, B. "Increasing Thigh Extension with Haptic Feedback Affects Leg Coordination in Young and Older Adult Walkers." Journal of Biomechanics, 2025.
5. Sharafian, E., Ghassabzadeh, M., & Taghvaeipour, A. "Trajectory Optimization of a Spot-Welding Robot in the Joint and Cartesian Spaces." International Journal of Robotics and Automation, 2023.
6. Sharafian, E., Taghvaeipour, A., & Ghassabzadeh, M. "Revisiting Screw Theory-Based Approaches in the Constraint Wrench Analysis of Robotic Systems." Robotica, 2022.